

Centrifugal Pump P-3, Plant Utilities

SAMPLE MAINTENANCE GUIDE FOR DEMONSTRATION.

Written for the JAMES demo. This is not an OEM document and must not be used on real equipment.

Contents: 1 Safety. 2 Description. 5 Troubleshooting. 6.3 Bearing and coupling inspection. 6.4 Alignment check. 6.5 Cavitation checks.

1 Safety

Apply lockout and tagout before any work that exposes rotating parts, the coupling or electrical terminals. Verify zero energy with a try-start from the local station.

Wear gloves and eye protection for all inspection work.

Bearing housings and the casing can be hot. Do not touch surfaces above the plant limit. Let the pump cool and measure before contact.

2 Description

Close-coupled end-suction centrifugal pump for chilled water.

Motor 7.5 kW, 2900 rpm. Drive-end bearing 6205-2RS. Flexible jaw coupling.

Normal vibration (sample value): below 4.5 mm/s RMS at the bearing housing.

Normal casing temperature (sample value): 35 to 50 C.

5 Troubleshooting

Symptom: high vibration with rising bearing temperature.

Likely cause: bearing wear. Go to 6.3 Bearing and coupling inspection.

Symptom: high vibration with normal temperature, often after coupling work.

Likely cause: misalignment. Go to 6.4 Alignment check.

Symptom: noise like gravel in the pump, discharge pressure fluctuating.

Likely cause: cavitation. Go to 6.5 Cavitation checks.

6.3 Bearing and coupling inspection

Procedure for suspected bearing wear.

Step 1: Review the vibration and temperature trend and confirm the fault.

Step 2: Isolate the motor at the breaker and apply your personal lock and tag.

Step 3: Verify zero energy with a try-start from the local station.

Step 4: Remove the coupling guard.

Step 5: Inspect the bearing housing for heat, noise and grease leakage and record findings.

If the bearing is damaged, replace bearing 6205-2RS and escalate to the senior technician.

6.4 Alignment check

Procedure for suspected misalignment.

Step 1: Review the vibration trend and recent coupling work.

Step 2: Isolate the motor at the breaker and apply your personal lock and tag.

Step 3: Verify zero energy with a try-start from the local station.

Step 4: Remove the coupling guard.

Step 5: Check coupling alignment with a straight edge and record findings.

6.5 Cavitation checks

Procedure for suspected cavitation.

Step 1: Review suction and discharge pressure readings.

Step 2: Check that the suction valve is fully open.

Step 3: Check the suction strainer differential pressure.

Step 4: Record findings and escalate if pressure stays unstable.